Eric Van Grinsven

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Education

University of California San Diego Sep. 2021 – Mar. 2023
M.S. in Computer Engineering | GPA: 3.76/4.0 *La Jolla, CA*

University of California San Diego Sep. 2018 – Jun. 2021
B.S. in Computer Engineering, Minor in Mathematics | GPA: 3.67/4.0 *La Jolla, CA*

Research Experience

Auditing for Multifairness Notions Aug. 2026 – Present
Independent Research

- Investigating conditions under which worst-group multiaccuracy error approximates distance to the set of multiaccurate predictors.

Memory Complexity of Testing Graph Properties Nov. 2024 – Sep. 2026
Research under Prof. Sumegha Garg, Rutgers University

- Jointly proved an $\widetilde{\Omega}(n/s)$ semi-random query lower bound for s-bit algorithms testing bipartiteness or detecting large cliques, using information-theoretic techniques.
- Contributed to the correctness proof of a clique-detection algorithm and extended its analysis to graphs with nonuniform vertex degrees.
- Adapted a random-walk bipartiteness tester to the semi-random query model.

Clique detection through random walks Mar. 2021 – Jun. 2021
Independent Study under Prof. Russell Impagliazzo, UC San Diego

- Studied papers on lower bounds for clique detection using random walks, drawing on random graph theory and the probabilistic method; attended research-team meetings.

Teaching Experience

University of California San Diego Sep. 2021 – Mar. 2023
Graduate Teaching Assistant, Computer Science and Engineering *La Jolla, CA*

- **Courses:** Design and Analysis of Algorithms, Discrete Mathematics, Introduction to Computational Theory, Operating Systems.
- Led discussion sections and office hours; supported course forums, grading, exam proctoring, and assignment review.
- Created supplementary instructional materials, including a Jupyter notebook connecting random graphs to problems in the algorithms course.

Honorable Mention for Excellence in Teaching (2021–22 academic year).

University of California San Diego Jan. 2020 – Jun. 2021
Undergraduate CSE Tutor *La Jolla, CA*

- **Courses:** Design and Analysis of Algorithms, Discrete Mathematics.
- Held office hours and supported course forums, grading, and exam proctoring.

Honorable Mention for Excellence in Teaching (2020–21 academic year).

AmeriCorps Sep. 2017 – Mar. 2018
Academic Coach *Eldon, MO*

- Tutored high school students in math and led classes and review sessions when teachers were absent.
- Ran after-school tutoring and support programs, providing consistent academic guidance and encouragement to students facing challenges at school and at home.

Professional Experience

Splunk Jun. 2025 – Present
Software Engineer *Durham, NC*

- Lead development of a highly available PostgreSQL-backed service, supporting reliable database infrastructure for internal services.
- Mentored two new grad engineers through design guidance, code reviews, walkthroughs, and technical documentation.

Bloomberg Apr. 2023 – Jun. 2025
Software Engineer *New York, NY*

- Developed backend systems for fixed-income electronic trading, including redesigning FIX message routing and parsing infrastructure serving more than 100 trading firms.
- Modernized C++ infrastructure processing more than two million daily runs and axes during trading hours, improving performance by 30%.

Robinhood Jun. 2022 – Sep. 2022
Software Engineer Intern *Menlo Park, CA*

- Built infrastructure connecting marketing tools with experimentation and A/B-testing data, saving approximately 20 hours of recurring product-management work per week.

Walmart Global Technology Jun. 2021 – Aug. 2021
Software Engineer Intern *Remote*

- Improved data integrity and reduced risk of data loss in Walmart's corporate badge-enablement system.

Selected Academic Projects

Non-Stationary Bandit Learning Aug. 2024 – Oct. 2024
Independent Study

- Reproduced phased exploration and exploitation from Ghosh et al.'s DNCB algorithm for two agents in Python with non-stationary arm rewards.

Iterative List Decoding Nov. 2022
Final presentation, DSC 291: Coding Theory, UC San Diego

- Co-presented Justesen et al.'s iterative list-decoding method, explaining syndrome-guided bit flipping, tree-based search, and decoding guarantees.

Blind Deconvolution May 2022
Final project, ECE 273: Convex Optimization and Applications, UC San Diego

- Implemented Ahmed et al.'s convex blind-deconvolution method; reproduced recovery experiments and tested its performance against a non-blind baseline and violations of its assumptions.

Analysis of Internet Traffic Through Major US Backbone Links Feb. 2021 – Mar. 2021
Final project, CSE 221: Graduate Operating Systems, UC San Diego

- Collaborated on a Python and tcpdump pipeline to analyze CAIDA packet traces from three U.S. backbone links.